

Demonstrate to me that you know the material. Use your words.

(1 pt E.C.) 1. What is your favorite kind of donut?

(5 pts) 2. List three anthropogenic effects that can induce alternate stable states in seagrass beds.

(10 pts) 3. You did a case study on the Line Islands. Within this island group, Kingman and Kiritimati differed greatly by top predator abundance and successful coral recruitment. Tell how each of these differed between the two islands and how eutrophication contributed to these differences.

- (5 pts) 4. A study showed that hydrothermal vent populations were not connected between the northern and southern regions of the East Pacific Rise. Predict how population connectivity between these two regions might be affected by ocean warming. Justify your prediction.
- (5 pts) 5. Write the equilibrium equation for the bicarbonate buffering system.
- (10 pts) 6. Using the equation from question 5, explain how addition of atmospheric CO_2 is causing the loss of aragonite in the oceans.

- (10 pts) 7. Crabs in a salt marsh can have a negative indirect interaction with *Spartina*, through snails. Explain the interaction, tell whether the interaction is trait- or density-mediated, and tell why.

- (10 pts) 8. For this course, you wrote a conservation plan for a specific ecosystem. For that ecosystem, describe two ecosystem services that will be affected negatively by either warming or acidification of the oceans. Describe each service, and then tell specifically how warming or acidification will alter the service.

(15 pts) 9. Diagram and describe the mutualistic mesograzer hypothesis. Your diagram should include two trophic levels above the mesograzers. If the top trophic level has low abundance, tell whether the seagrasses will be abundant or not, and why.

(5 pts) 10. Tell whether the effect that the top trophic level you diagrammed for question 9 has on seagrasses is a TMII or a DMII and why.